Yiyu Zhu

Full-Stack Software Engineer

zhuyy8612@gmail.com · GitHub portfolio · Remote contracts · 20 hours/week · China UTC+8

Builds Node.js and Python applications with persistent state, failure recovery and reproducible tests. Supports AI application integration at China Telecom.

Stack: JavaScript / Node.js 24 · Python · SQL / SQLite · HTTP APIs · pandas / openpyxl · Streamlit · pytest · GitHub Actions

RunLedger Durable Webhook Delivery

Independent project · Node.js 24 · SQLite · Browser console · Nov 2026 · Source and verification

Delivers AI job completion events and exposes retries, receiver outages and dead-letter recovery in an operator console.

- Commits each event and delivery together. Canonical payloads deduplicate under a stable idempotency key; changed payloads return HTTP 409. WAL-aware backups restore queue history and leases to fresh files without overwriting existing data.
- Claims tasks with expiring leases and fences stale worker completions. Retries transient failures with jitter and respects the receiver's Retry-After minimum; replay preserves identity and attempt generations.
- Signs raw messages with HMAC and uses a persistent receiver inbox to deduplicate receipts under at-least-once delivery. Verified redelivery after a lost acknowledgment and restart.
- Adds whole-ledger status/search, cursor pagination and read-transaction metrics for queue health and replay-aware p50/p95/p99 latency. Thirty-five Node.js scenarios verify queue and HTTP behavior.
- Reproduced two-process claim competition and worker termination. Sent 200 local requests in batches of 20 with two workers: 100 repeats deduplicated, 100 unique events delivered and 100 inbox receipts. Linux/Windows and browser checks run in CI.

Selected Projects

OrderCheck Order Data Quality Workbench Validates money with Decimal, isolates conflicting order groups and exports a seven-sheet audit workbook. Seventy tests cover rules, provenance, UI/CLI behavior and malformed KPI input.

PriceWatch Historical Monitoring Serializes predecessor lookup and snapshot commit in SQLite. Validates complete alert policies before network or output side effects. Fifty-three tests include competing writers, rollback and CLI failures.

LeadReview and API Contract Lab Separates AI extraction, deterministic scoring and human review. Generated n8n code is tested without module globals. HTTP/JSON/SSE regression tooling passes 18 self-tests; lead rules have 17 explicit assertions.

Experience

China Telecom Innovation Product Operations Center · Jun 2026-Present

- Support AI application integration: reproduce recognition, configuration and event-delivery failures; adjust conditions and retest to isolate the cause.
- Record expected and actual behavior, reproduction steps and retest results; maintain configuration checks for project support.

Education and Languages

City University of Hong Kong · MSc, Electronic Information Engineering · Sep 2025-Jun 2026

Sun Yat-sen University · BEng, Aerospace Engineering · Sep 2021-Jun 2025

Native Mandarin · English CET-6 505 · English-medium postgraduate study